Finite-Time Adaptive Linear Neuron Control for Torque Ripple Suppression of Permanent Magnet Synchronous Motors

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***Abstract*—Permanent Magnet Synchronous Motors (PMSMs) are widely employed in industrial applications for high-performance motion control. However, their speed regulation is often challenged by disturbances such as torque ripple and parameter uncertainties, which may induce speed oscillations and degrade motion smoothness and system reliability. To address these issues, this paper proposes a finite-time control (FTC) strategy integrated with an Adaptive Linear Neuron (ADALINE) network to achieve robust speed regulation under both periodic and aperiodic disturbances. The ADALINE is trained online via a Lyapunov-based finite-time adaptation law, enabling accurate compensation of unknown disturbance profiles without extensive controller tuning. The proposed method ensures finite-time convergence of the speed tracking error, thereby enhancing both transient performance and steady-state precision. Experimental results demonstrate the proposed controller effectively suppresses speed ripple and ensures fast recovery under varying speed and load conditions, validating its robustness and broad applicability for high-performance PMSM speed control.**

***Index Terms*—Adaptive linear neuron (ADALINE), finite-time control (FTC), speed ripple suppression.**

I. Introduction

Permanent Magnet Synchronous Motors (PMSMs) are widely used in applications such as electric vehicles, industrial automation, and precision machining, owing to their high efficiency, power density, and excellent controllability [1]. However, achieving precise speed regulation in these systems remains a significant challenge due to their susceptibility to torque ripple, which can considerably degrade overall performance. In PMSM drives, the primary cause of torque ripple is the inherent cogging effect between the rotor and stator [2]. Additionally, magnetic asymmetries and inaccuracies in current sampling also contribute to these torque fluctuations [3], [4]. These oscillations propagate to the rotor speed, reducing steady-state accuracy and often causing mechanical vibrations and acoustic noise, thereby undermining the reliability and quality of motion control.

To address these challenges, various control strategies have been proposed to mitigate the adverse effects of torque ripple and enhance speed regulation performance. Among these, approaches based on the internal model principle have received considerable attention. For instance, [5] presents a repetitive control scheme combined with an extended state observer (ESO) to compensate for periodic disturbances, aperiodic uncertainties, and nonlinearities in the plant. However, repetitive control exhibits poor adaptability to frequency variations, which limits its effectiveness under changing operating conditions. Moreover, the combination of ILC and conventional PI control has been employed to suppress periodic torque ripple. However, this approach remains highly sensitive to model uncertainties, and its performance can be severely degraded in the presence of aperiodic perturbations [6]. Resonant control techniques have also been widely used in both controller and observer designs, such as in Proportional-Integral-Resonant (PIR) controllers [7] and Generalized Internal Model-based Extended State Observers (GIESOs) [8], to suppress specific harmonic components. However, these methods often suffer from implementation challenges, as their stability depends heavily on precise parameter tuning, and addressing multiple harmonics typically requires the inclusion of several resonant modules, which increases system complexity. Moreover, in

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PMSM applications, it is essential to maintain high dynamic performance under frequently changing operating conditions, such as variable speeds and load fluctuations. However, for resonant control methods, sudden non-periodic disturbances often lead to distortion in the output of the resonant terms, which in turn induce speed oscillations and result in prolonged transient periods.

To overcome these problems, the Adaptive Linear Neuron (ADALINE) can be considered a promising candidate due to its simple structure, low computational burden, and ease of implementation [9]. In [10], ADALINE is employed to extract current harmonics and generate corresponding compensation voltages. Furthermore, [11] integrates ADALINE into an ESO structure to address the deadtime effect. These studies demonstrate that ADALINE provides strong suppression capability for specific harmonic orders. However, the use of the least mean square (LMS) or gradient descent adaptation law leads to only asymptotic stability, which may limit the dynamic performance of the system.

In this paper, an ADALINE is integrated into a finite-time control (FTC) framework to simultaneously reject both periodic and aperiodic disturbances. In systems with multiple harmonic components, the ADALINE can adaptively learn the disturbance characteristics by updating its network parameters online, thereby avoiding the need for complex controller parameter tuning. Moreover, by employing a finite-time Lyapunov-based adaptation law for weight updates, the network is capable of accurately approximating and compensating various types of disturbances while ensuring excellent dynamic performance. As a result, smooth and robust speed regulation of PMSM drives can be achieved under diverse operating conditions.

II. System Modeling and Preliminaries

The mechanical dynamics of PMSMs can be described by the following equation:

$$\dot{\omega}_m = \frac{T_e - B\omega_m - T_L}{J} \tag{1}$$

where ω_m is the mechanical angular speed, T_e is the electromagnetic torque, J is the moment of inertia, B is the friction coefficient, T_L is the load torque.

In practical PMSM drive systems, speed regulation is frequently influenced by various disturbances, including parameter uncertainties such as fluctuations in rotational inertia and time-varying external loads. Moreover, periodic torque ripple introduces speed oscillations, further deteriorating the smoothness of motor operation. Taking into account these disturbances, as well as the tracking errors introduced by the inner torque control loop, the dynamic behavior of the speed loop can be modeled as follows:

$$\begin{aligned}\dot{\omega}_m &= b\left[T_e^* - \left(T_e^* - T_e + B\omega_m + T_L + T_{rip}\right)\right] \\ &= b_n T_e^* - \left(b_n - b\right) T_e^* - b\left(T_d + T_{rip}\right) \\ &= b_n T_e^* + d_s\end{aligned} \tag{2}$$

where $b = 1/J$ and b_n are the control gain and its nominal value, $T_d = T_e^* - T_e + B\omega_m + T_L$ is the lumped disturbance torque, T_{rip} denotes the torque ripple, and d_{to} represents the total disturbance.

According to the torque ripple expressions summarized in [7], the total disturbance d_s, which consists of both periodic and aperiodic components, can be further represented as:

$$d_s = d_{ac} + d_{dc} = \sum_{i=1}^{\infty}\left[A_i \sin\left(n_i\theta_m\right) + B_i \cos\left(n_i\theta_m\right)\right] + d_{dc} \tag{3}$$

where d_{ac} and d_{dc} represent the total periodic and aperiodic disturbances, θ_m denotes the mechanical rotor position, n_i is the harmonic order, A_i and B_i are the corresponding amplitudes.

A. Preliminaries

Lemma 1 ([12]): Consider the nonlinear system $\dot{x} = f(x), x(0) = x_0$ with $x \in R^n$, let $V(x)$ be a continuously differentiable positive-definite function. Suppose there exist constants $\zeta > 0$, $\eta > 0$ and $0 < \kappa < 1$ such that

$$\dot{V}(x) \le -\zeta V^{\kappa}(x) + \eta. \tag{4}$$

then the system is semiglobally practically finite-time stable. Specifically, for all $t \ge T$ and $0 < \varrho \le 1$,

$$V^{\kappa}(t) \le \frac{\eta}{(1-\varrho)\zeta},$$

with settling time

$$T = \frac{1}{(1-\kappa)\varrho\zeta}\left[V^{1-\kappa}(0) - \left(\frac{\eta}{(1-\varrho)\zeta}\right)^{\frac{1-\kappa}{\kappa}}\right].$$

Lemma 2 (Young's Inequality [13]): For any $a, b \in R^n$, and for $\mu, \nu > 1$ such that $\frac{1}{\mu} + \frac{1}{\nu} = 1$, the following inequality holds for any $\epsilon > 0$:

$$ab \le \epsilon\frac{a^{\mu}}{\mu} + \frac{b^{\nu}}{\nu\epsilon^{\nu/\mu}}. \tag{5}$$

Lemma 3 ([14]): For $0 < \gamma < \frac{\gamma_1}{\gamma_2} < 1$, where γ_1 and γ_2 are odd integers, and $\tilde{x} = x - \hat{x}$, the following inequality holds:

$$\tilde{x}\left(x - \tilde{x}\right)^{\gamma} \le \iota_1 x^{1+\gamma} - \iota_2 \tilde{x}^{1+\gamma}, \tag{6}$$

where $\iota_1 = \frac{1}{1+\gamma} - \frac{2^{\gamma-1}}{1+\gamma} + \frac{2^{2\gamma-\gamma^2-1}}{(1+\gamma)^2} + \frac{\gamma}{(1+\gamma)^2}$, and $\iota_2 = \frac{2^{\gamma-1} - 2^{(1+\gamma)(\gamma-1)}}{1+\gamma} > 0$.

III. Design of the Finite-time Controller using ADALINE

A. Finite-time Control Law

Using ω_m^* denotes the reference value of ω_m, the speed tracking error can be defined as

$$e_s = \omega_m^* - \omega_m \tag{7}$$

Based on (2), the derivative of (7) can be expressed as

$$\dot{e}_s = \dot{\omega}_m^* - \dot{\omega}_m = \dot{\omega}_m^* - b_n T_e^* - d_s \tag{8}$$

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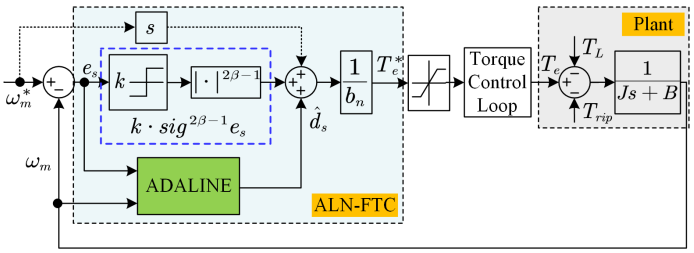

Fig. 1. Block diagram of the proposed speed regulation system.

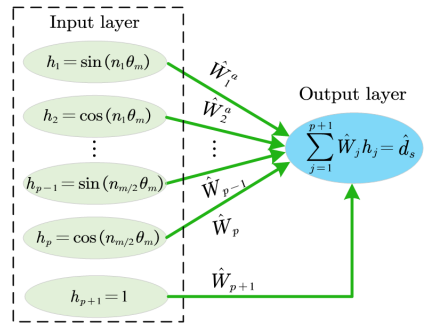

Fig. 2. Structure of the proposed ADALINE.

Combining (8) with the finite time theorem, the torque reference is obtained as

$$T_e^* = \frac{\dot{\omega}_m^* + k_p sig^{2\beta-1} e_s - d_s}{b_n} \tag{9}$$

where $k_P > 0$ and $sig^{\beta} e_s = |e_s|^{\beta} sign\,(e_s)$.

Generally, the actual values of the system's disturbances are unknown. Hence, it is a common practice to substitute it by the estimated component $\hat{d}_s$. In this way, (9) is modified as

$$T_e^* = \frac{\dot{\omega}_m^* + k_p sig^{2\beta-1} e_s - \hat{d}_s}{b_n} \tag{10}$$

To avoid overcurrent in the practical system, the reference torque saturation limit is usually applied as

$$T_{e,sat}^* = \begin{cases} T_{e,\max}, & T_e^* \geqslant T_{e,\max}^* \\ T_e^*, & T_{e,\min}^* < T_e^* < T_{e,\max}^* \\ T_{e,\min}, & T_e^* \leqslant T_{e,\min}^* \end{cases} \tag{11}$$

where $T_{e,sat}^*$, $T_{e,\max}^*$ and $T_{e,\min}^*$ are the saturated torque reference, the maximum torque reference, and the minimum torque reference, respectively.

B. Disturbance Estimation Based on ADALINE

To eliminate steady-state speed deviations and suppress oscillations, the estimated disturbance $\hat{d}_s$ must closely match its actual value. However, the disturbance d_s often consists of multiple harmonic components and may change rapidly over time, making accurate observation a considerable challenge. Given the inherent periodic nature of d_s, the ADALINE is employed due to its ability to approximate arbitrary nonlinear relationships and perform online learning. By embedding ADALINE within the proposed finite-time control framework, the overall ALN-FTC speed regulation system is constructed, as depicted in Fig. 1.

Equation (3) can be rewritten in the following matrix form:

$$d_s = [A_1, B_1 ... A_N, B_N, d_{dc}] \cdot [\sin(n_1\theta_m), \cos(n_1\theta_m) ... 1]^T \tag{12}$$

From this matrix multiplication representation, the architecture of ADALINE, which consists solely of input and output layers with a linear mapping between them, is illustrated in Fig. 2, and its input vector is defined as follows:

$$h = \left[\sin(n_1\theta_m), \cos(n_1\theta_m), ..., \cos\left(n_{\frac{p}{2}}\theta_m\right), 1\right]^T \tag{13}$$

where h is the input vector and p is the number of sinusoidal components utilized in the input layer neurons.

In the input layer, each pair of sinusoidal and cosinusoidal signals with a common period is employed to observe disturbances associated with that period. The ADALINE then estimates the overall disturbance by linearly combining these basis signals using its weight vector $\hat{W} = \left[\hat{W}_1, \hat{W}_2 ... \hat{W}_{p-1}, \hat{W}_p, \hat{W}_{p+1}\right]^T$, Consequently, the ADALINE output $\hat{d}_s$ is given by

$$\hat{d}_s = \sum_{j=1}^{p+1} \hat{W}_j h_j = \hat{W}^T h \tag{14}$$

where j is a positive integer.

Similarly, the actual disturbance can also be expressed in the NN form as below:

$$d_s = W^{*T} h + \varsigma \tag{15}$$

where W^* is the optimum weight vector, ς denotes the ADALINE approximation error, which is bounded by a positive constant such that $|\varsigma| < \bar{\varsigma}$.

The disturbance estimation error is thus defined as

$$\tilde{d}_s = d_s - \hat{d}_s = \tilde{W}^T h + \varsigma \tag{16}$$

where $\tilde{W} = W^* - \hat{W}$ and $\tilde{d}_s$ are the estimation error of weight vector and system disturbance, respectively.

With the objective of minimizing speed tracking error, the following adaptation law for weight vector training is designed as below:

$$\dot{\hat{W}} = -k_w \left(e_s h + \sigma \hat{W}^{2\beta-1}\right) \tag{17}$$

where k_a is the adaptation gain, σ is a positive constant.

With this online adaptive mechanism, the weight vector is continuously adjusted to capture both periodic and aperiodic disturbance components. Owing to its parallel computational structure, ADALINE enables rapid response in both transient and steady-state conditions.

By combining equations (8), (10), and (16), the resulting dynamics of the speed tracking error is formulated as:

$$\dot{e}_s = -k_p sig^{2\beta-1} e_s - \tilde{W}^{\mathrm{T}} h - \varsigma \tag{18}$$

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C. Stability Analysis of the Closed-loop System

Theorem 1: Consider the PMSM system governed by (2). Under the finite-time control law given in (10), in conjunction with the ADALINE adaptive update rule (17), the speed tracking error is guaranteed to converge to a small vicinity of zero within a finite time. Furthermore, all signals in the closed-loop system remain uniformly bounded throughout the operation.

Proof: Consider the following Lyapunov candidate function:

$$V = \frac{1}{2}e_s^2 + \frac{1}{2k_a}\tilde{W}^T\tilde{W} \tag{19}$$

Taking the time derivative of V and using the error dynamics from (18), we obtain:

$$\dot{V}_0 = -k_p|e_s|^{2\beta} - e_s\tilde{W}^T h - e_s\varsigma - \frac{1}{k_a}\tilde{W}^T\dot{\hat{W}} \tag{20}$$

To handle the cross term $e_s\varsigma$, we apply Young's inequality, which yields:

$$e_s\varsigma \leq \frac{1}{2\beta}|e_s|^{2\beta} + \frac{2\beta-1}{2\beta}\bar{\varsigma}^{\frac{2\beta}{2\beta-1}} \tag{21}$$

Substituting both the adaptive update law (17) and the inequality (21) into (20), we can further bound $\dot{V}_0$ as:

$$\dot{V}_0 \leq -\left(k_p - \frac{1}{2\beta}\right)|e_s|^{2\beta} + \sigma\tilde{W}^T\hat{W}^{2\beta-1} + \frac{2\beta-1}{2\beta}\bar{\varsigma}^{\frac{2\beta}{2\beta-1}} \tag{22}$$

For the nonlinear term $\sigma_2\tilde{W}^{\mathrm{T}}\hat{W}^{2\alpha-1}$, we invoke *Lemma 3*, which provides the following inequality:

$$\begin{aligned}\sigma\tilde{W}^T\hat{W}^{2\beta-1} &= \sigma\tilde{W}^T\left(W^* - \tilde{W}\right)^{2\beta-1} \\ &\leq -l_1\sigma\left(\tilde{W}^T\tilde{W}\right)^{\beta} + l_2\sigma\left(W^{*T}W^*\right)^{\beta}\end{aligned} \tag{23}$$

where $l_1 = \frac{2^{2\beta-2}-2^{(2\beta)(2\beta-2)}}{1+\beta}$ and $l_2 = \frac{1}{2\beta} - \frac{2^{2\beta-2}}{2\beta} + \frac{2(2\beta-1)-(2\beta-1)^2-1}{(2\beta)^2}$.

By substituting the bound in (23) into inequality (22), we further obtain:

$$\begin{aligned}\dot{V}_0 \leq &-\left(k_p - \frac{1}{2\beta}\right)|e_s|^{2\beta} - l_1\sigma\left(\tilde{W}^T\tilde{W}\right)^{\beta} \\ &+ \frac{2\beta-1}{2\beta}|\bar{\varsigma}|^{\frac{2\beta}{2\beta-1}} + l_2\sigma\left(W^{*T}W^*\right)^{\beta}\end{aligned} \tag{24}$$

Now, define the constant $\lambda = \min\left\{2k_p - \frac{1}{\beta}, (2k_a)^{\alpha}\, l_1\sigma\right\}$. Then inequality (24) can be reformulated into a more compact form as:

$$\dot{V} \leq -\lambda V^{\beta} + \delta \tag{25}$$

where $\delta = \frac{2\beta-1}{2\beta}|\bar{\varsigma}|^{\frac{2\beta}{2\beta-1}} + l_2\sigma\left(W^{*\mathrm{T}}W^*\right)^{\beta}$ is a positive constant.

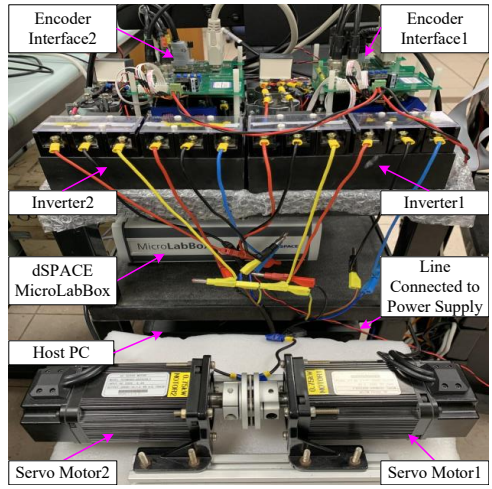

Fig. 3. Configuration of the test bench.

TABLE I
PARAMETERS OF THE EXAMINED IPMSM

Symbol	Quantity	Symbol	Quantity
Polepair numbers p_n	4	Rated power P_N	0.75 (kW)
Stator resistance R_s	1.1 (Ohm)	D axis inductance L_d	5.7 (mH)
Motor system inertia J	4.44×10^{-4} (kgm^2)	Q axis inductance L_q	5.7 (mH)
Rated torque T_N	2.4 (Nm)	Current limit I_{smax}	9 (A)

Referring to *Lemma 1*, the proposed closed-loop system is SGPFS. Specifically, there exists a constant T such that for all $t \geqslant T$ the following holds:

$$V(t) \leq \left(\frac{\delta}{(1-\varepsilon)\lambda}\right)^{\frac{1}{\beta}}, \qquad \forall\, t \geq T, \tag{26}$$

where $0 < \varepsilon \leq 1$ Moreover, the settling time T is bound by

$$T^* = \frac{1}{(1-\beta)\varepsilon\lambda}\left[V^{1-\beta}(0) - \left(\frac{\delta}{(1-\varepsilon)\lambda}\right)^{\frac{1-\beta}{\beta}}\right]. \tag{27}$$

IV. Experimental Verification

A. System Configuration

The specifications of the evaluated PMSM are detailed in TABLE I, with the test bench schematic illustrated in Fig.3. The system employs two inverters, each incorporating an Intelligent Power Module (IPM) and Hall sensors, to drive the PMSM and the load motor separately. A programmable DC power supply provides the bus voltage, which is set to 150V. The control algorithms are deployed on a dSPACE MicroLabBox, with both the sampling and control frequencies configured at 10kHz.

Three control schemes are evaluated in the experiment:

C1: PI Controller. The speed control law in the Laplace domain is expressed as $T_e^*(\mathrm{s}) = \left(k_p + \frac{k_i}{\mathrm{s}}\right)e_s(\mathrm{s})$, where the proportional gain and integral gain are set as $k_p = 200$ and $k_i = 10000$.

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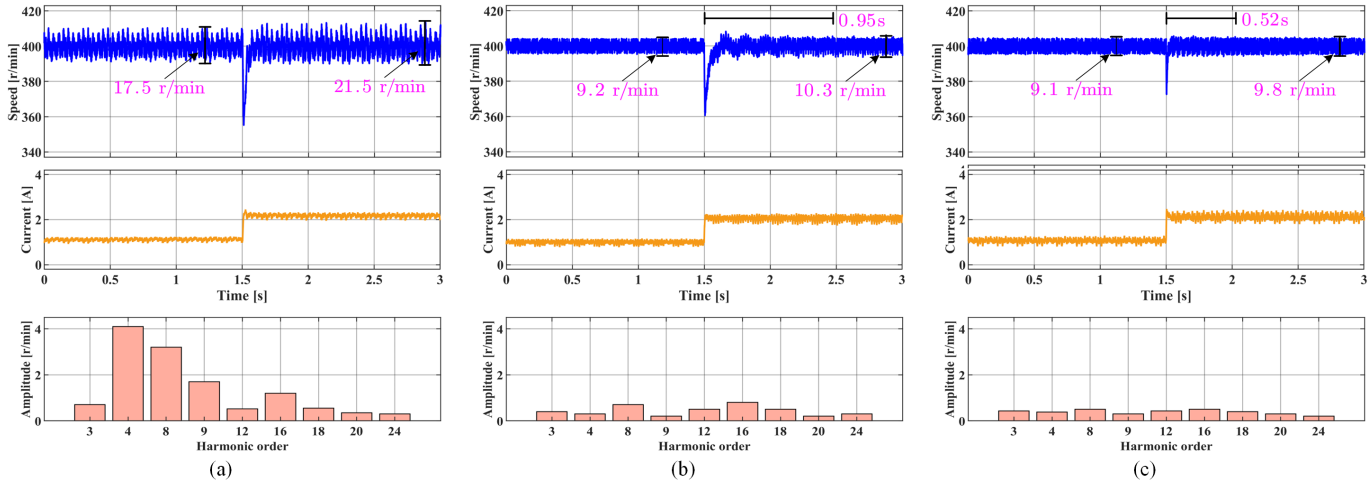

Fig. 4. Experimental results under load torque varying from 25% to 50%: (a) C1: PI Controller (b) C2: PIR Controller (c) C3: the Proposed Method.

C2: PIR Controller [7]. The control law extends the PI structure by incorporating resonant compensators and is given by $T_e^*(\mathrm{s}) = \left(k_p + \frac{k_i}{\mathrm{s}}\right) e_s(\mathrm{s}) + \sum_{i=1}^{N} \frac{2k_r \mathrm{s}}{\mathrm{s}^2 + (i\omega_m)^2} e_s(\mathrm{s})$, where $k_r = 1000$ denotes the resonant gain. According to the harmonic spectrum analysis under the C1 controller (discussed in the following section), four resonant terms corresponding to harmonic orders $i = 4, 8, 9, 16$ are included to target the dominant harmonic components.

C3: Proposed Controller (10). The controller gain is set as $k = 200$. For the ADALINE, the parameters are configured as $k_a = 10000$ and $\sigma = 0.01$.

B. Experimental Results under Load Torque Change

The variable-load experiment is conducted, in which the load torque is increased from 20% to 50% of its rated value. The experimental results under the three control strategies, including the speed response, q-axis current and FFT analysis of the speed tracking error, are shown in Fig. 4.

When the conventional PI controller is applied, as illustrated in Fig. 4(a), the speed response exhibits significant ripple, which becomes more pronounced under the heavier load. The FFT results reveal that the speed signal contains a wide range of harmonic components, reflecting the fact that cogging torque, influenced by manufacturing imperfections and magnetic non-idealities, consists of multiple order harmonics. Among these, the 4th, 8th, 9th, and 16th harmonics are dominant. Additionally, the sudden load increase causes a substantial speed drop exceeding 40r/min.

To suppress these dominant harmonics, the PIR controller is employed, and its performance is illustrated in Fig. 4(b). It can be seen that, despite the increased load, the speed ripple is significantly reduced. FFT analysis confirms that the targeted harmonics are effectively attenuated to minimal levels. However, abrupt aperiodic disturbances still cause a noticeable speed drop and slow down the recovery process. Moreover, because the resonant terms are designed to target specific harmonic frequencies, the controller becomes sensitive to transient deviations from the nominal operating point. As a result, oscillations arise during the dynamic response, which prolong the transient period to nearly one second following the load change.

In contrast, the ADALINE-based method exhibits robust and consistent performance under both load conditions, as shown in Fig. 4(c). Its ability to suppress harmonic disturbances is comparable to that of the PIR controller. Additionally, the speed drop is significantly smaller and the recovery is faster, owing to the finite-time control design. Specifically, when the sudden load is applied, the ADALINE network quickly estimates the aperiodic disturbance. Through continuous online parameter adaptation, it also effectively approximates and compensates for the periodic disturbances over time. As a result, the transient period after load application is notably short, lasting for approximately 0.5 seconds.

C. Experimental Results under Speed Change

Figure 5 presents the experimental results for a speed transition from 200 r/min to 400 r/min under the three different control strategies. As shown in Fig.5(a), under the conventional PI controller, significant speed harmonics are observed at both 200 r/min and 400 r/min, with the low-speed condition exhibiting more severe harmonic content. This is mainly because PMSMs inherently attenuate high-frequency disturbances more effectively at higher speeds, while at lower speeds, high-order harmonics are less attenuated. In addition, the response to the speed command change is relatively slow and accompanied by a noticeable overshoot.

When the PIR controller is applied, as illustrated in Fig. 5(b), the speed harmonics under different speeds are effectively suppressed. However, the dynamic performance deteriorates compared to the PI controller. In particular, obvious speed oscillations occur following the speed change, along with a larger overshoot. These oscillations mainly arise from the resonant terms, since the sudden increase in speed tracking error distorts the resonant output. Consequently, the transient period is extended, and the system takes approximately 1.08 seconds to stabilize.

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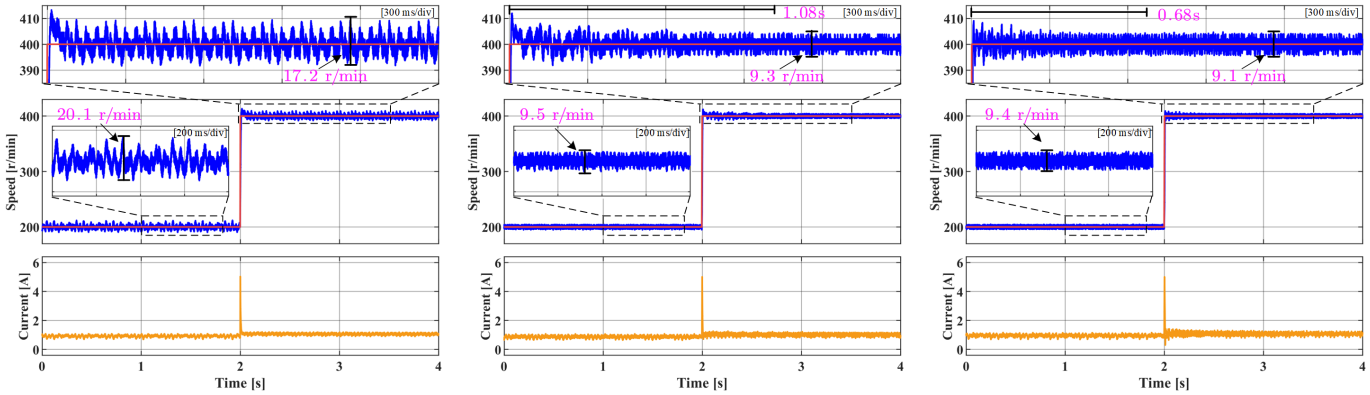

(a) (b) (c)

Fig. 5. Experimental results under speed varying from 200 r/min to 400 r/min: (a) C1: PI Controller (b) C2: PIR Controller (c) C3: the Proposed Method.

In comparison, when the proposed method is applied, Fig. 5(c) demonstrates substantial improvement in dynamic performance. The overshoot is minimal, and the transient period lasts only around 0.68 seconds, which can be attributed to the fast convergence property of the proposed scheme. Specifically, as the operating speed changes, the ADALINE network retrains its weight parameters to adapt to the new conditions. Benefiting from the parallel computation structure and the FTC-based adaptive law, this re-optimization process is completed efficiently, enabling the system to quickly regain stability. Furthermore, the steady-state harmonic suppression performance at different speeds is comparable to that of the PIR controller, highlighting the ADALINE's capability to accurately approximate disturbances online and maintain consistent performance.

V. Conclusion

In this work, a finite-time control scheme incorporating the ADALINE was developed to achieve smooth and robust speed regulation for PMSM drives. The proposed method employs a neural weight adaptation law derived from Lyapunov stability theory, enabling accurate real-time estimation and compensation of both periodic and aperiodic disturbances. Finite-time stability of the closed-loop system is theoretically guaranteed, ensuring fast convergence of the speed tracking error. Experimental validation on a 0.75 kW PMSM platform demonstrates that the controller significantly suppresses speed ripple and delivers fast, stable dynamic responses across a wide range of operating conditions. These results highlight the effectiveness and broad applicability of the proposed ADALINE-based finite-time control strategy in high-performance PMSM applications.

References

[1] G. Wang, M. Valla, and J. Solsona, "Position sensorless permanent magnet synchronous machine drives—a review," *IEEE Transactions on Industrial Electronics*, vol. 67, no. 7, pp. 5830–5842, 2020.

[2] M. Aydin and M. Gulec, "Reduction of cogging torque in double-rotor axial-flux permanent-magnet disk motors: A review of cost-effective magnet-skewing techniques with experimental verification," *IEEE Transactions on Industrial Electronics*, vol. 61, no. 9, pp. 5025–5034, 2014.

[3] M. S. Rafaq, W. Midgley, and T. Steffen, "A review of the state of the art of torque ripple minimization techniques for permanent magnet synchronous motors," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 1, pp. 1019–1031, 2024.

[4] C. Zhao, Y. Zuo, H. Wang, Q. Hou, S. Zhu, and C. H. Lee, "Torque ripple suppression of pmsm speed regulation system using neural network," in *2022 IEEE 7th Southern Power Electronics Conference (SPEC)*, 2022, pp. 1–6.

[5] A. H. M. Sayem, Z. Cao, and Z. Man, "Model free eso-based repetitive control for rejecting periodic and aperiodic disturbances," *IEEE Transactions on Industrial Electronics*, vol. 64, no. 4, pp. 3433–3441, 2017.

[6] J. Liu, H. Li, and Y. Deng, "Torque ripple minimization of pmsm based on robust ilc via adaptive sliding mode control," *IEEE Transactions on Power Electronics*, vol. 33, no. 4, pp. 3655–3671, 2018.

[7] C. Xia, B. Ji, and Y. Yan, "Smooth speed control for low-speed high-torque permanent-magnet synchronous motor using proportional-integral-resonant controller," *IEEE Transactions on Industrial Electronics*, vol. 48, no. 10, pp. 2650–2658, 2012.

[8] Y. Zuo, J. Zhu, W. Jiang, S. Xie, X. Zhu, W.-H. Chen, and C. H. T. Lee, "Active disturbance rejection controller for smooth speed control of electric drives using adaptive generalized integrator extended state observer," *IEEE Transactions on Power Electronics*, vol. 38, no. 4, pp. 4323–4334, 2023.

[9] C. Zhao, Y. Zuo, H. Wang, H. Cao, and C. H. T. Lee, "Smooth speed control for electric drives based on adaptive rbf neural network method," *IEEE Transactions on Power Electronics*, pp. 1–10, 2025.

[10] T. Qiu, X. Wen, and F. Zhao, "Adaptive-linear-neuron-based dead-time effects compensation scheme for pmsm drives," *IEEE Transactions on Power Electronics*, vol. 31, no. 3, pp. 2530–2538, 2016.

[11] C. Zhao, Y. Zuo, K. Yu, H. Wang, T. Li, and C. H. T. Lee, "Adaptive linear neuron-based active disturbance rejection control system for current harmonics suppression of pmsm drives," in *2024 27th International Conference on Electrical Machines and Systems (ICEMS)*, 2024, pp. 1861–1866.

[12] F. Wang, B. Chen, X. Liu, and C. Lin, "Finite-time adaptive fuzzy tracking control design for nonlinear systems," *IEEE Transactions on Fuzzy Systems*, vol. 26, no. 3, pp. 1207–1216, 2018.

[13] H. Deng and M. Krstic, "Output-feedback stochastic nonlinear stabilization," *IEEE Transactions on Automatic Control*, vol. 44, no. 2, pp. 328–333, 1999.

[14] Y. Wang, Y. Song, M. Krstic, and C. Wen, "Fault-tolerant finite time consensus for multiple uncertain nonlinear mechanical systems under single-way directed communication interactions and actuation failures," *Automatica*, vol. 63, pp. 374–383, 2016.

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